

Frozen scores → candidate policies → certified release policy → common evaluation

Frozen predictor

gene-GO score matrix s_{ig}
per-call source sets $\mathcal{R}_{ig}$
(annotated module members)

never retrained

Random split of the panel

ranking fold $\mathcal{T}$ (10%)
certification fold $\mathcal{C}$ (60%)
evaluation fold $\mathcal{E}$ (30%)
exchangeable by construction

GoDag path

global thresholds
 $\lambda_1 > \lambda_2 > \dots > \lambda_l$
one cutoff for every gene
tied scores move together

GoCo path (this paper)

same grid, plus within each block
partial admission $\pi_{j,q}$ of the
affected genes ordered by u_i
(Eq. 6; fitted on $\mathcal{T}$ only)
nested, $\mathcal{T}$ -measurable

Certification on $\mathcal{C}$

$\hat{R}_k, \hat{\sigma}_k$ over certification genes
 $p_k = \Phi\{(\hat{R}_k - \alpha)/(\hat{\sigma}_k/\sqrt{m})\}$
fixed sequence: stop at first $p_k > \delta$

Prop. 1: $\Pr\{R(\hat{\pi}) > \alpha\} \leq \delta$
for any $\mathcal{T}$ -measurable ordering

Evaluation on $\mathcal{E}$

common TruePath truth for
every method

gene-level FDP (controlled)
supported genes / terms
calls, supported fraction